Protein Metabolism

Second stage

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Introduction

Proteins are the most abundant organic molecules in cells, representing about (50%) of their dry weight. They are found in every part of every cell and there are many kinds of proteins, each specialized for different biological function.

A large number of different proteins have been isolated in pure crystalline form.

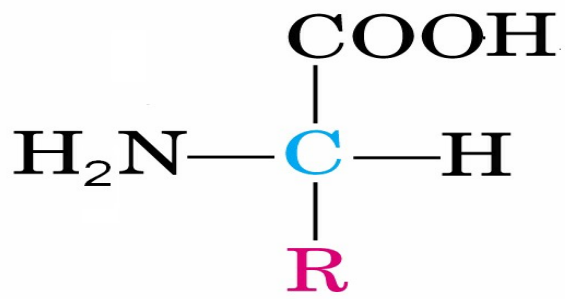
All contain Carbon, Hydrogen, Oxygen and Nitrogen.

Most proteins contain (S) and certain proteins contain additional elements particularly Phosphorous, Iron, Zinc and Copper.

The molecular weight of proteins are very high (macro molecules), but on hydrolysis they all give a group of simple organic compounds of low molecular weight called alpha amino acids.

Alpha amino acids are building blocks of proteins and contain at least one -COOH and one -NH2 groups, but differs one from the other by the chemical nature of the side chain

(R group)

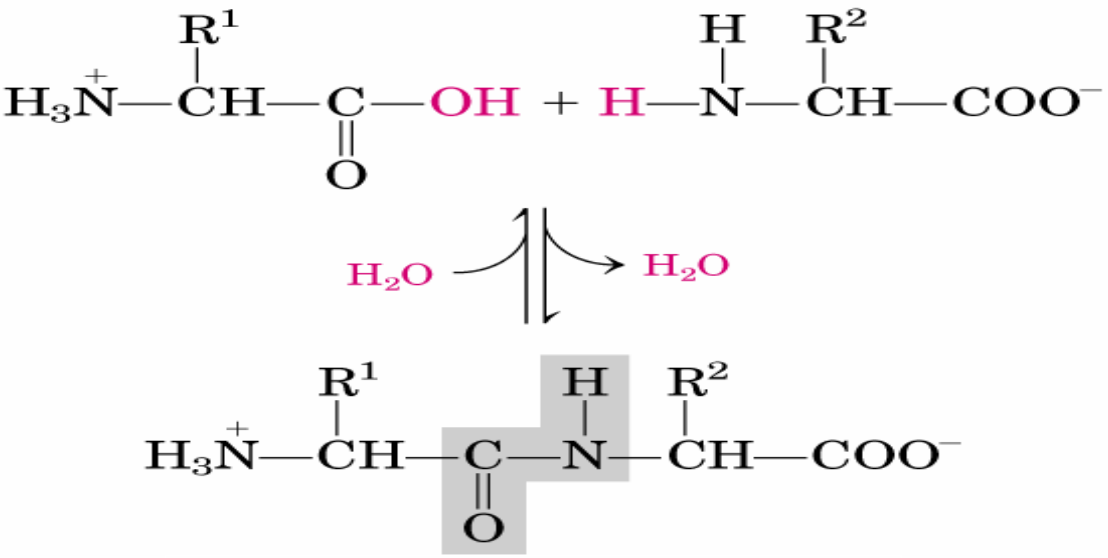


General structure of amino acid

Many amino acids are found in nature

(> 300), but only 20 different alpha amino acids are commonly found as building blocks of proteins.

Amino acids are linked together to form peptides, polypeptides through an amide linkage called peptide bond .



Peptide bond

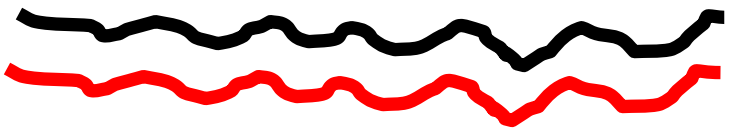
Polypeptides are also macromolecules and may contain up to(100) amino acids .

Some proteins contain only one polypeptide chain, others contain(2) or more polypeptide chains. Proteins contain

(2) or more polypeptide chains are known as oligomeric proteins and their component chains are called subunits or protomers.

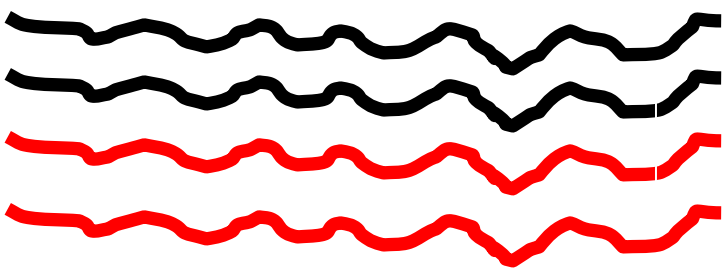
Examples of oligomeric proteins

1-Insulin



Alpha chain = 21 amino acids
Beta chain = 30 amino acids

2-Hb



Alpha chain = 141 amino acids
Alpha chain = 141 amino acids
Beta chain = 146 amino acids
Beta chain = 146 amino acids

DIGESTION AND ABSORPTION OF PROTEINS

The amino acids ingested by human mainly in the form of proteins which can not directly enter the metabolic pathways. So, the ingested proteins must undergo first digestion (hydrolysis) in the G.I.T. and this process consists essentially in the hydrolysis of proteins to their constituent amino acids and small peptides in the presence of proteolytic enzymes of G.I.T.

In the stomach, where hydrochloric acid denatures protein to make the peptide bonds more available to the actions of enzymes. HCl also converts pepsinogen to the active enzyme pepsin, which begins the process of breaking down proteins into smaller polypeptides and some amino acids.

Most of the protein digestion occurs in the small intestine, where pancreatic proteases reduce polypeptides to shorter chains, tripeptides, dipeptides, and amino acids.

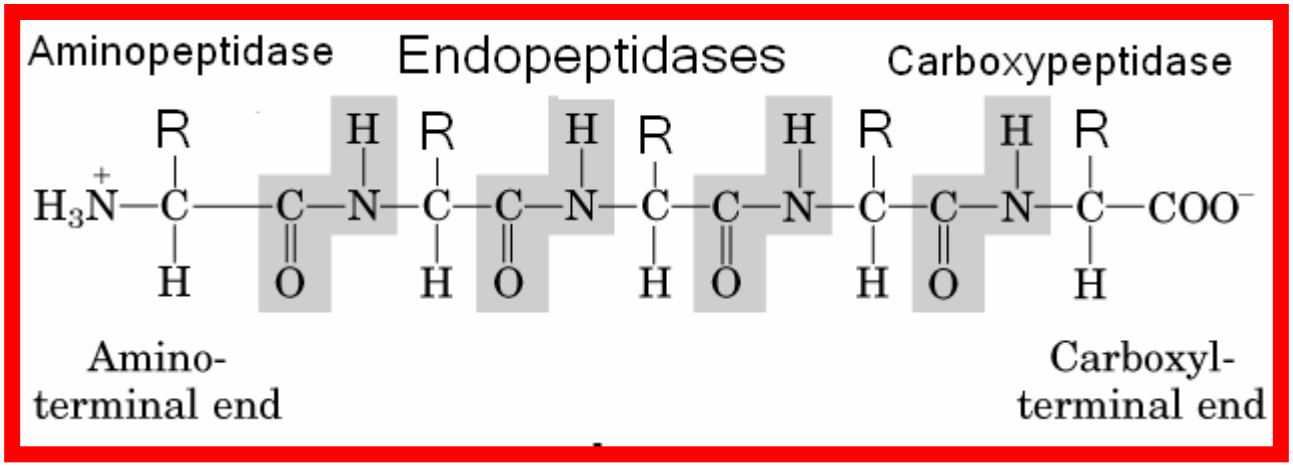
The proteolytic enzymes of small intestine are of two types :

1-Endopeptidases: include trypsin and chymotrypsin, hydrolyze the inner peptide bonds of peptides or proteins.

2-Exopeptidases: subdivided into two subtypes:

A-Aminopeptidases: hydrolyze the peptide bond of the N-terminal amino acid.

B-Carboxypeptidases: hydrolyze the peptide bond of the C-terminal amino acid .


Aminopeptidase
Endopeptidases
Carboxypeptidase
Amino-terminal end
Carboxyl-terminal end

As a result of the action of exo and endopeptidases, proteins are converted to free amino acids and short peptides contain not more than(8) amino acids. (octapeptides)

This mixture is then absorbed actively by small intestine and here the hydrolysis is completed by dipeptidases. From intestinal mucosa the mixture of amino acids transported to portal blood and then to liver. The liver removes certain proportion of amino acids to supply its own needs and at the same time adds other amino acids which are synthesized by the liver into the blood.

The tissue proteins are also continually undergoing catabolism and adding also amino acids to the blood.

So, the blood amino acids arise from: absorption, synthesis by liver and tissue catabolism, and this represents what is called (amino acid pool).

Each tissue of the body takes from this pool the specific amino acids in proper proportion to synthesize the proteins required for growth, maintenance and proper functions . Another portion of the amino acids of the pool is used for the synthesis of non protein nitrogenous compounds (N.P.N.) such as choline, creatine—etc. which are also essential for certain tissues.

Amino acids in excess are catabolized in liver to form ammonia and keto acids.

Ammonia is converted into urea in liver and excreted in urine. The liver converted keto acids into new amino acids by (amination or transamination) or to glucose by (gluconeogenesis). Except CO_2 most of the waste products (catabolites) of protein catabolism are excreted in urine such as ammonia, urea, creatinine. Since nitrogen is the most characteristic and constant product of protein and protein catabolism, its estimation is necessary to determine the overall process of protein metabolism and consequently nitrogen balance. To determine nitrogen balance, the nitrogen contents of the food eaten and that of waste products excreted in urine and feces in a given time are obtained and there will be(3) possibilities :

1-If the nitrogen content of foods exceeds that excreted, this means that the body is taking nitrogen principally as tissue proteins and this condition is called positive nitrogen balance, such as those occur in cases of infants, children, pregnancy.

2-If the nitrogen excreted is equal to that of the food taken, the condition is called nitrogen balance in which catabolism = anabolism, and this is usually the case of adults .

3-If the nitrogen excreted exceeds that of food eaten, the condition is called negative nitrogen balance such as those occur in cases of illness, starvation, protein deficiency, surgery, advanced cancer.

CLASSIFICATION OF PROTEINS

Different methods are proposed for protein classification:

1-Depending on their size and shape:

A-Globular proteins: characterized by the presence of peptide chains folded or coiled in a very compact manner. They are soluble in water with a classic hydrophilic surface .

Examples: albumin, globulin, insulin, myoglobin.

B-Fibrous proteins: they are elongated thread like, water insoluble and play structural roles, found in teeth, tendons and skin. They have helical structures

Examples: collagen, keratin

2- Depending on their chemical composition:

A-Simple proteins :they give on hydrolysis only alpha amino acids. Examples: albumin, globulin .

B-Complex (conjugated) proteins: they contain in addition to alpha amino acids other materials. Examples: haemoglobin, lipoproteins, nucleoproteins, mucoproteins .

C-Derived proteins: they are derived from simple or complex proteins .

Examples: peptone, co-agulated proteins

(co-agulated albumin).

3- Depending to their functions:

A-Structural proteins: they provide structural components such as collagen and keratin.

B-Contractile proteins: they are involved in the movement of muscles such as myosin and actin.

C-Transport proteins: they carry essential substances throughout the body such as hemoglobin for oxygen, lipoproteins for lipids, ceruloplasmin for copper, transferrin for iron.

D-Storage proteins: they store nutrients such as casein of milk, ferritin store Iron.

E-Hormones: they regulate body metabolism and nervous system such as insulin, growth hormone.

F-Enzymes: they catalyze biochemical reactions in the cells such as sucrase, trypsin.

G-Protection proteins: they recognize and destroy foreign substances such as immunoglobulins .

DAILY PROTEIN REQUIREMENT

The number of proteins in the diet which is necessary to supply an optimum quantity of nitrogen for the maintenance of nitrogen balance is known as daily protein requirement and this depends not on quantity but quality of proteins, because the needs of human body is mainly for specific amino acids which are not synthesized by the human body and their absence in the diet leads to negative nitrogen balance. These amino acids are known as essential (indispensable) amino acids. Essential amino acids are needed for:

1-Synthesis of new body proteins during growth.

2-For continuous regeneration of tissue proteins in adult .

3-For synthesis of hormones, enzymes, purines, creatine-- etc.

***1 gm /Kg body weight / day**

ESSENTIAL AMINO ACIDS	NON-ESSENTIAL AMINO ACIDS
Valine	Glycine
Leucine	Alanine
Isoleucine	Aspartic acid
Arginine	Aspargine
Histidine	Glutamic acid
Lycine	Glutamine
Threonine	Serine
Methionine	Cysteine
Tryptophan	Tyrosine
Phenyl alanine	Proline

PROTEIN STORAGE

There are no evidences of the existence within the human body for storage of proteins like the well-known ability of the body to store carbohydrates and fats.

The reserve proteins which are used during periods of fasting and starvations or protein deficiency appear to be given from tissue themselves.

Classification of amino acids

A-The (20) amino acids in proteins classified according to the (R) group, according to their chemical structure:

1- Aliphatic amino acids;

Glycine

Alanine

Valine

Leucine

Isoleucine

2-Amino acids containing hydroxyl (-OH) group;

Serine

Threonine

3-Amino acids containing thiol (-SH) group;

Cysteine

Methionine

4-Acidic amino acids and its amides;

Aspartic acid

Aspargine

Glutamic acid

Glutamine

5-Basic amino acids;

Arginine

Lysine

Histidine

6-Hetercyclic amino acids

(Containing imino -NH- group);

Proline

7-Aromatic amino acids;

Phenyl alanine

Tyrosine

Tryptophan

B-Depending on their polarity:

1-Polar amino acids : they are soluble in water because them (R) are poplar such as - OH,-SH ,-COO, -NH2

2-Non-polar amino acids : they are insoluble in water ,because their (R) are non-polar .

C-Depending whether they are building blocks of proteins or not they are divided into :

1-Building blocks of proteins ,such as : Alanine, Tryptophan ,Lysine----etc.

2-Not building blocks of proteins such as : Ornithine, Citrulline ------etc.

Selenocysteine; the twenty first amino acid

In addition to the 20 standard amino acids found in proteins, a twenty first amino acid has been discovered and shown to be an essential residue in several enzyme system, selenocysteine occurs at the active site of several enzymes, that includes:

- Glutathione peroxidase
- Thioredoxin redactase
- Glycine reductase
- Formate dehydrogenase
- Deiodinase that converts thyroxine to triiodothyronine

Selenoprotein P: a glycoprotein containing 10 selenocysteine residues, found in blood. It has an antioxidant function, and its concentration falls in selenium deficiency.

Peptides

Amino acids bind together by peptide bonds, to produce, di, tri, tetrapeptides and so on, 2-20 amino acids called **oligopeptides.**

21-50 amino acids binds together called **polypeptides**

More than 50 amino acids regards as **proteins.**

Bioactive Peptides (BP)

Bioactive Peptides (BP) have been defined as specific protein fragments that have a positive impact on body functions or conditions and may influence health. Currently, more than 1500 different BP have been reported in a database named **‘Biopep’**

BP are organic substances formed by amino acids joined by peptide bonds.

BP play important roles in the metabolic functions of living organisms and, consequently, in human health. They display hormone or drug-like activities and can be classified based on their mode of action as antimicrobial, anti thrombotic, antihypertensive, immunomodulatory, mineral binding, hypoglycemic agent, anticancer and antioxidative.

Example of bioactive peptide hydrolysis from proteins in

Moringa oleifera



100 g of leaf of Moringa	
Calories	13.4 calories
Protein	**2.57 grams**
Fats	0.294 grams
Carbohydrates	1.74 grams
Fiber	0.42 grams
Calcium	38.8 milligrams
Iron	0.84 milligrams
Magnesium	8.82 milligrams

PROTEIN in moringa oleifera hydrolyzed to peptides that useful in medicine for that called

bioactive peptides

Medical benefits bioactive peptides extract in moringa oleifera

- Cancer prevention
- Treating anemia
- Treating asthma
- Antibacterial properties
- Boosting immunity
- antioxidants

Example of bioactive dipeptide;

Carnosine (*beta*-alanyl-L-histidine)

Is a dipeptide molecule, made up of the amino acid's beta-alanine and histidine. It is highly concentrated in muscle and brain tissues.

Carnosine is naturally produced by the body in the liver from beta-alanine and histidine. There are no plant-based sources of carnosine. Carnosine is readily available as a synthetic nutritional supplement.

Carnosine can chelate divalent metal ions.

Physiological effects of carnosine

•pH buffer

Carnosine has a pK_a value of 6.83, making it a good buffer for the pH range of human muscles.

•Anti-oxidant

Carnosine has been shown to scavenge reactive oxygen species (ROS) as well as alpha-beta unsaturated aldehydes formed from peroxidation of cell membrane fatty acids during oxidative stress.

- **Antiglycating**

Carnosine acts as an antiglycating agent, reducing the rate of formation of advanced glycation end-products (substances that can be a factor in the development or worsening of many diseases, such as diabetes, atherosclerosis, chronic kidney failure, and Alzheimer's disease

- **Geroprotective**

Carnosine is considered as a geroprotector. Carnosine can increase the human fibroblasts, as well as appearing to reduce the telomere shortening rate. Carnosine may also slow aging through its anti-glycating properties (chronic glycolysis is speculated to accelerate aging).

METABOLISM OF PROTEINS

This is essentially the metabolism of amino acids and can be divided into:

A-Anabolism:

1-Anabolism of non-essential amino acids.

2-Amino acids as biosynthetic precursors : certain amino acids are essential precursors of a variety of important biomolecules such as: nucleotides, nucleotide co-enzymes, hormones

(adrenaline, noradrenaline) ,neurotransmitters

(dopamine, serotonin, GABA =Gamma Amino Butyric Acid),

glutathione ----etc.

B-Catabolism :

1-Fate of NH2 group :

a-Deamination.

b-Transamination.

c-Urea cycle

2-Catabolism of carbon skeleton

C-Clinical correlation :

1-Metabolic disorders of urea cycle

2-Metabolic disorders of amino acid metabolism

A-Anabolism

1-Anabolism of non-essential amino acids:

Amino acids metabolism is a central importance in the metabolism of all organisms because they are the precursors of proteins which are building blocks of all living systems .

Different organisms vary considerably in their ability to synthesize amino acids and in the form of nitrogen used in this purpose . Vertebrates are not able to synthesize all the common amino acids, for example man can make only (10) of the (20) amino acids required as building blocks of proteins. The remainder (essential amino acids) must be obtained from diet .

Higher animals can use ammonium ion as the nitrogen source for the synthesis of non-essential amino acids, but they are unable to use NO_2, NO_3 or atmospheric nitrogen.

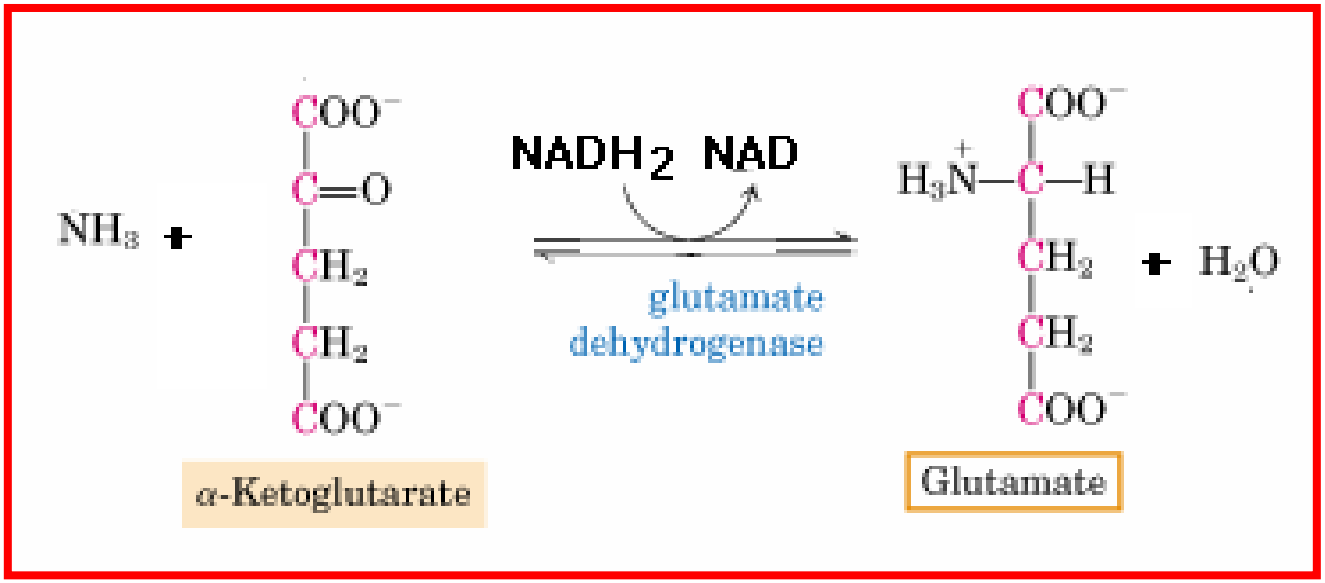
Higher plants can make all the amino acids required for protein synthesis and can use ammonia,NO_2, NO_3 and atmospheric nitrogen as a source of nitrogen.

The (20) different amino acids are synthesized by (20) different multi enzyme sequences, some of which are very complex .

Examples

1-Biosynthesis of glutamic acid

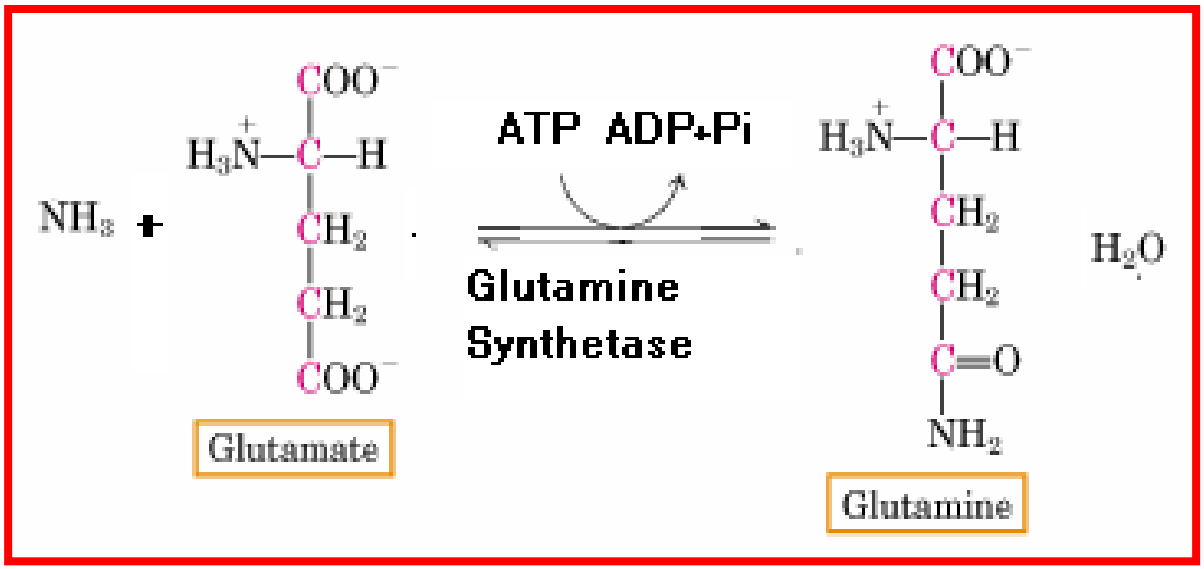
The main pathway for glutamate formation in most organisms is by amination of alpha ketoglutarate in the presence of glutamate dehydrogenase which has either NAD or NADP as a co-enzyme .



This reaction is fundamental importance in the biosynthesis of all amino acids, since in many species it's the principal pathway for the formation of alpha amino group directly from ammonia. In addition, the amino group of glutamate is transferred to various alpha keto acids to give the corresponding amino acids, so glutamic acid may be considered as universal amino acid or universal amino group donor .

2-Biosynthesis of glutamine

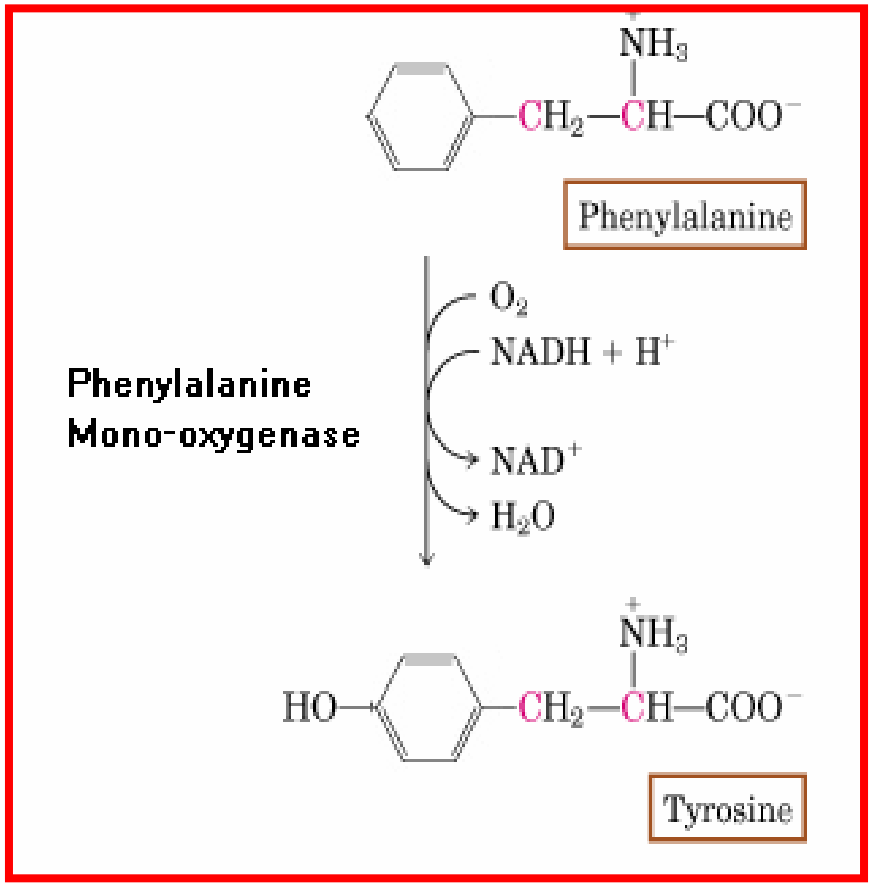
Glutamine is formed from glutamic acid by amination in the presence of glutamine synthetase



The enzyme glutamine synthetase plays an important role in the metabolism of amino acids, since glutamine is not only the non toxic transport form of ammonia, but also functions as an amino group donor in many biosynthetic reactions.

3-Biosynthesis of tyrosine

Its formed from the essential amino acid phenylalanine in a hydroxylation reaction catalyzed by phenylalanine mono-oxygenase



2-Amino acids as biosynthetic precursors:

Here we discuss the biotransformation of the carbon skeleton of amino acids, the amino acids, or a portion of their structures to important biomolecules.

1-Glycine: **it has an important role in the following anabolic processes :**

a-Biosynthesis of heme: the alpha carbon and nitrogen atoms of glycine are used in biosynthesis of the porphyrin portion of Hb.

b-Biosynthesis of purines: the complete molecule of glycine is used to form purine skeleton .

c-A constituent of glutathione: glycine is an important component of glutathione an important tripeptide that has an important role in metabolism .

d-Biosynthesis of creatine: the complete molecule of glycine is involved in the biosynthesis of creatine.

e-Conjugation with glycine :

Glycine conjugates with bile acids for example cholic acid forming glycocholic acid and with benzoic acid forming hippuric acid (Detoxification).

Clinical correlations

Glycine promotes insulin sensitivity by increasing glutathione synthesis, which combats oxidative stress, and may stimulate a gut hormone that enhances insulin's effect on glucose removal.

High plasma glycine levels are associated with improved insulin sensitivity, whereas low levels are seen in **type 2 diabetes**.

2-Serine : it's involved in:

a-Biosynthesis of brain sphingomyelins .

b-Biosynthesis of purine and pyrimidine bases .

3-Methionine : it's an important source of CH_3 group, It's the most important CH_3 group donor. The CH_3 group of methionine can be transferred into drugs contain pyridine or pyridine derivatives resulting in detoxification of these drugs .

4-Cysteine : it's involved in:

a-Biosynthesis of CoA .

b-It’s the precursor of the amino acid taurine that conjugates with bile acid cholic acid forming taurocholic acid.

5-Tryptophan: its precursor of the important neurotransmitter serotonin.

6-Phenylalanine and tyrosine: they are involved in:

a-Biosynthesis of the photoprotective pigment (melanin), the pigment of skin and hair.

b-Biosynthesis of catecholamines and thyroid hormones.

B-Catabolism

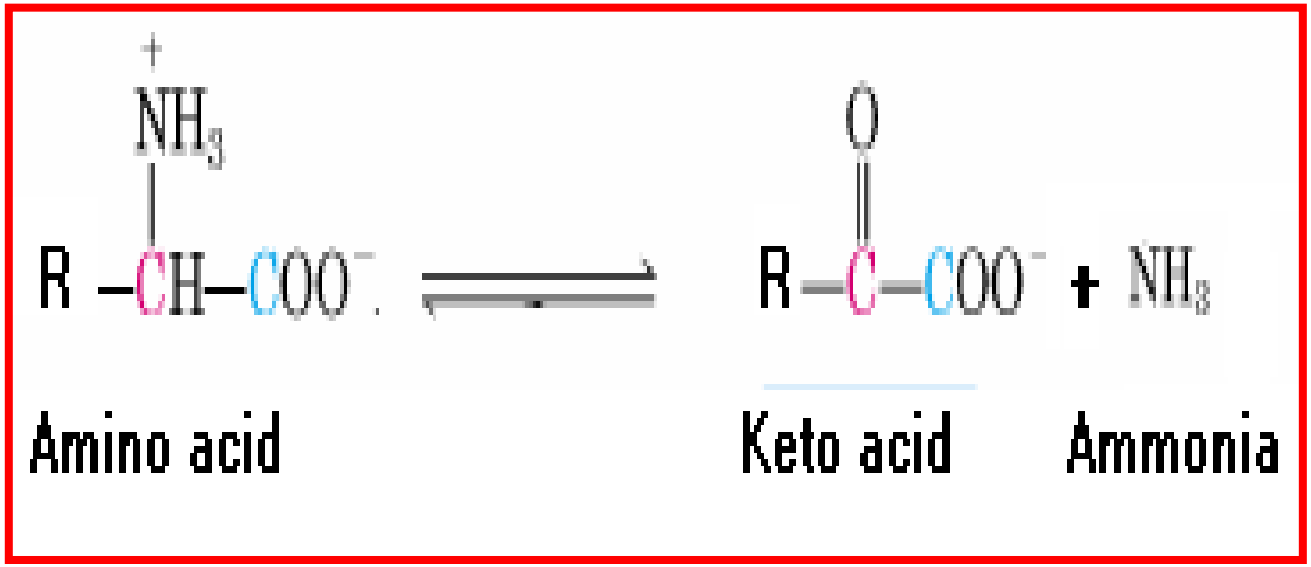
1-Fate of NH2 group :

a-Deamination: **it occurs in different tissues, but mainly in kidneys and liver and by this process many amino acids are converted to their corresponding ketoacids in the presence of specific amino acid oxidases which have either FMN or FAD as a co-enzyme.**

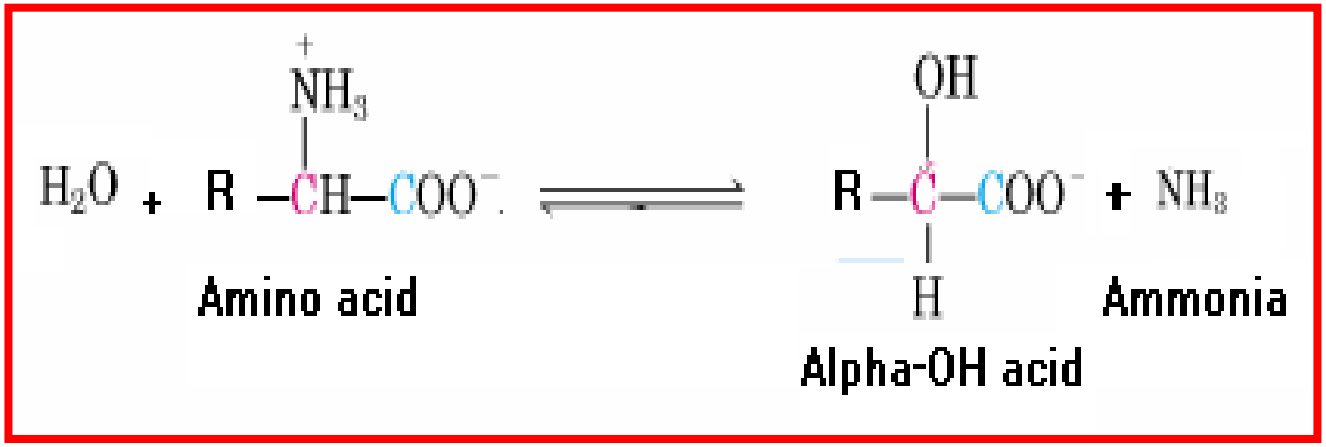
Generally, there are (3) types of deamination:

1-Oxidative deamination:

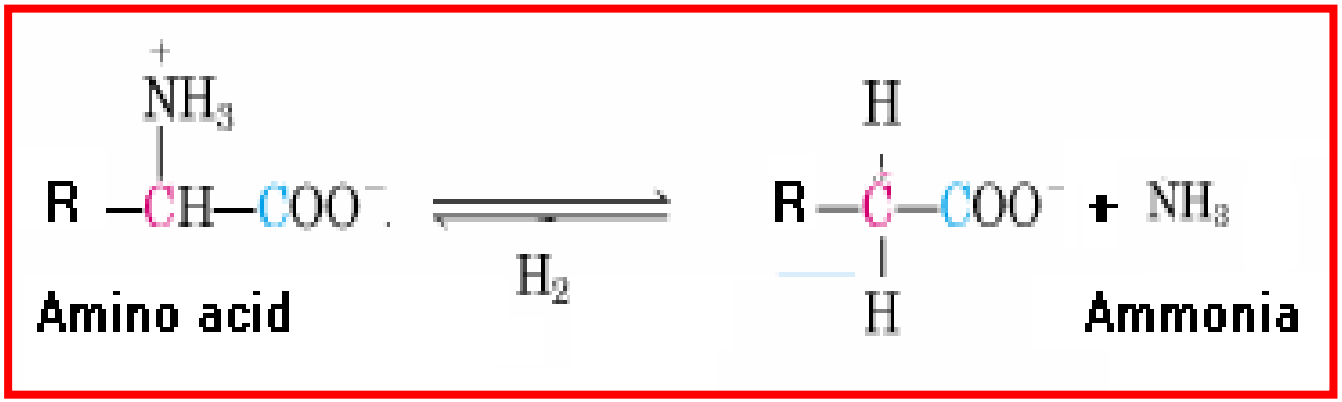
This is the principal and the most important type of deamination.



2-Hydrolytic deamination : its relatively less important



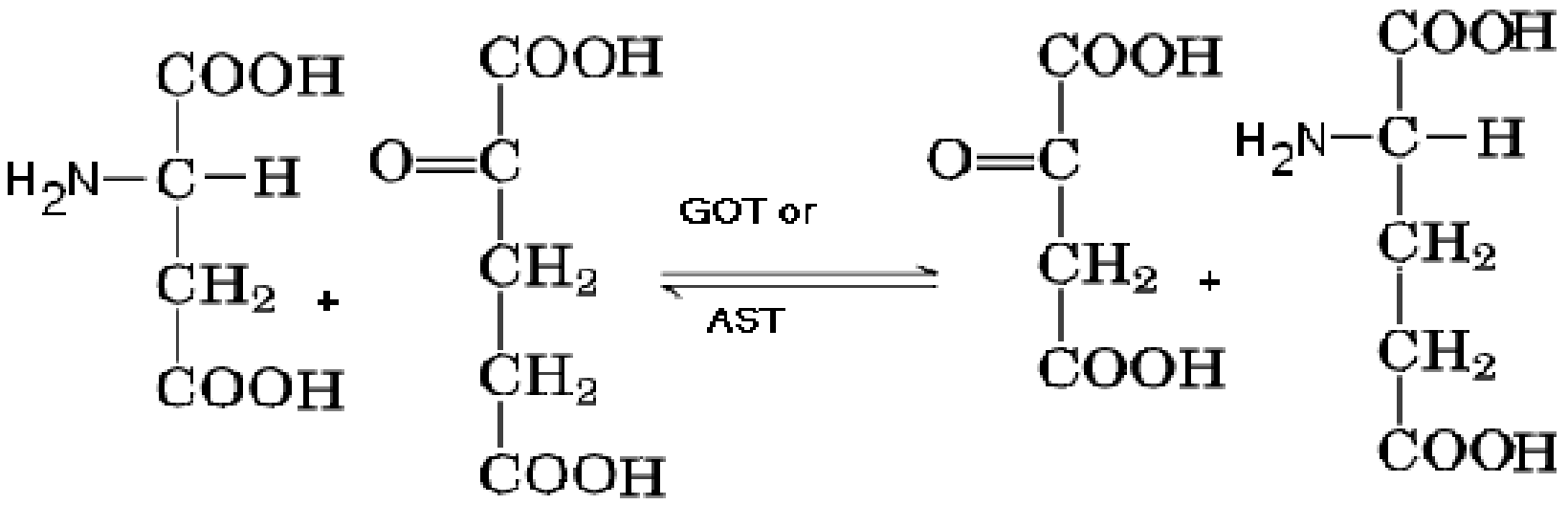
3-Reductive deamination: its also relatively less important



b-Transamination:

Is the process of combined deamination and amination (catabolism and anabolism) according to which the NH2 group of an amino acid is reversibly transferred to a keto acid in most cases alpha ketoglutarate.

These reactions are catalyzed by transaminases or amino transferases found in different tissues, but mainly in liver, kidney, heart and brain.



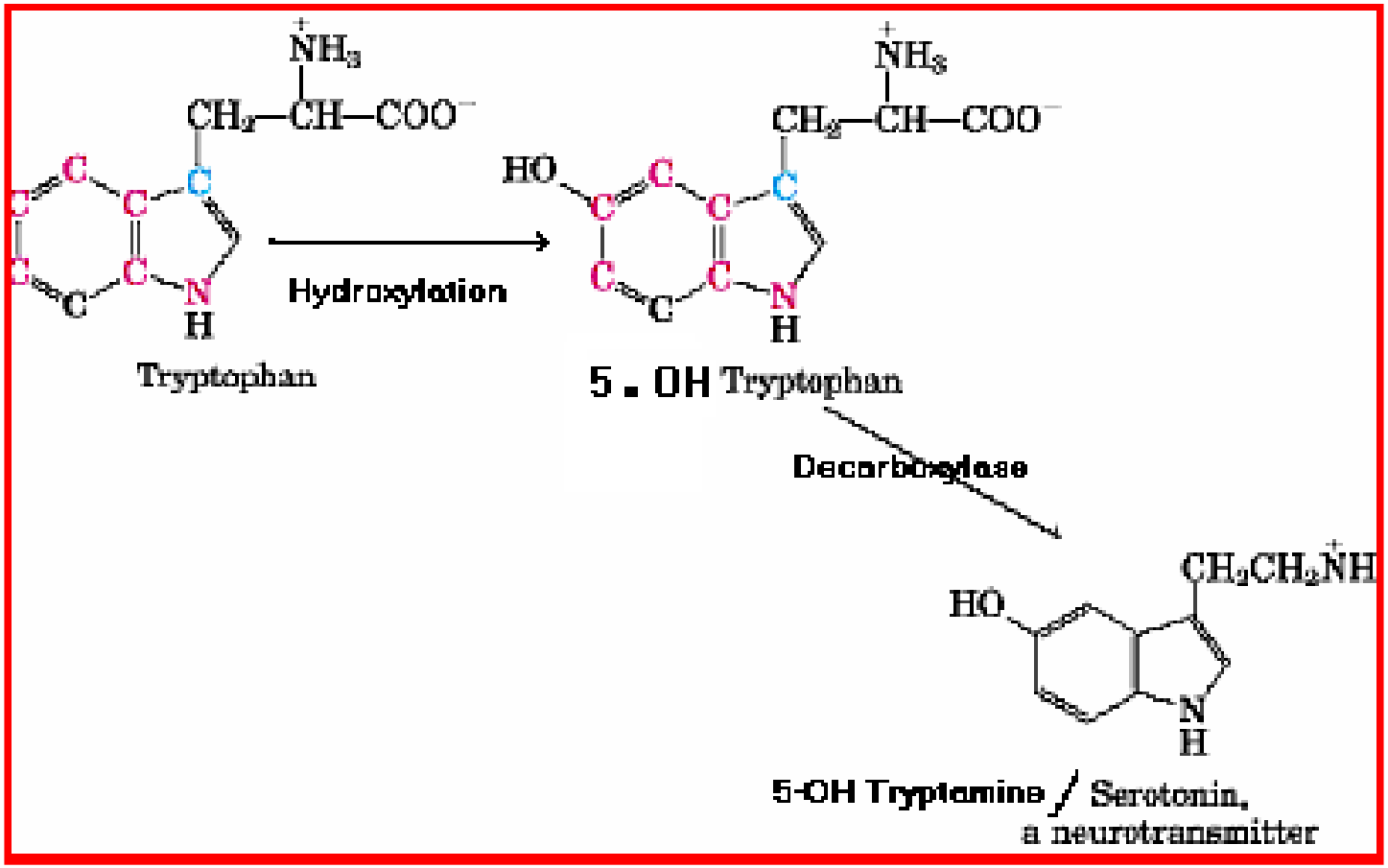
The resulting glutamic acid then undergoes oxidative deamination to form ammonia and the corresponding keto acid. Transamination process includes at least (12) amino acids: alanine, arginine, aspartic acid, aspargine, cysteine, lysine, leucine, isoleucine, valine, phenylalanine ,tyrosine and tryptophan.

The co-enzyme in transamination reaction is PLP which acts as NH2 carrier .

Decarboxylation:

Few amino acids undergo decarboxylation in animal tissues to give the corresponding primary amines which have also special biological importance.

Histidine —*histidine decarboxylase*→ **Histamine**


Tryptophan
Hydroxylation
5 - OH Tryptophan
Decarboxylase
5-OH Tryptamine / Serotonin,
a neurotransmitter

c-Urea cycle

In man ammonia is formed as a result of transamination and deamination of amino acids.

The formed ammonia may be excreted as such, but only in small amounts.

Certain amount of ammonia may be used for the biosynthesis of new amino acids by amination of ketoacids, mostly alpha ketoglutarate .

The majority of ammonia is converted to urea in a process known as urea cycle.

Urea synthesis through urea cycle

Under normal condition an adult excretes about (16.5) gm of nitrogen daily,

about (95%) is excreted in urine by the kidneys and the remaining (5%) in the stool.

The principal pathway of nitrogen excretion in human is as urea, synthesized in the liver, releases into the blood and cleared by the kidneys . Urea cycle includes (5) reactions :

1-Synthesis of carbamoyl-p-:

Condensation of one mole of CO_2, ammonia and phosphate derived from ATP to form carbamoyl phosphate.

This reaction is catalyzed by the enzyme carbamoyl phosphate synthetase of liver mitochondria. Mg and a dicarboxylic acid mainly N-acetyl glutamate is required in this reaction as enzyme activator .

2-Synthesis of citrulline :

This is done by transfer of carbamoyl moiety (portion) from carbamoyl phosphate to ornithine in the presence of ornithine transcarbamoylase of liver mitochondria .

3-Synthesis of argininosuccinic acid :

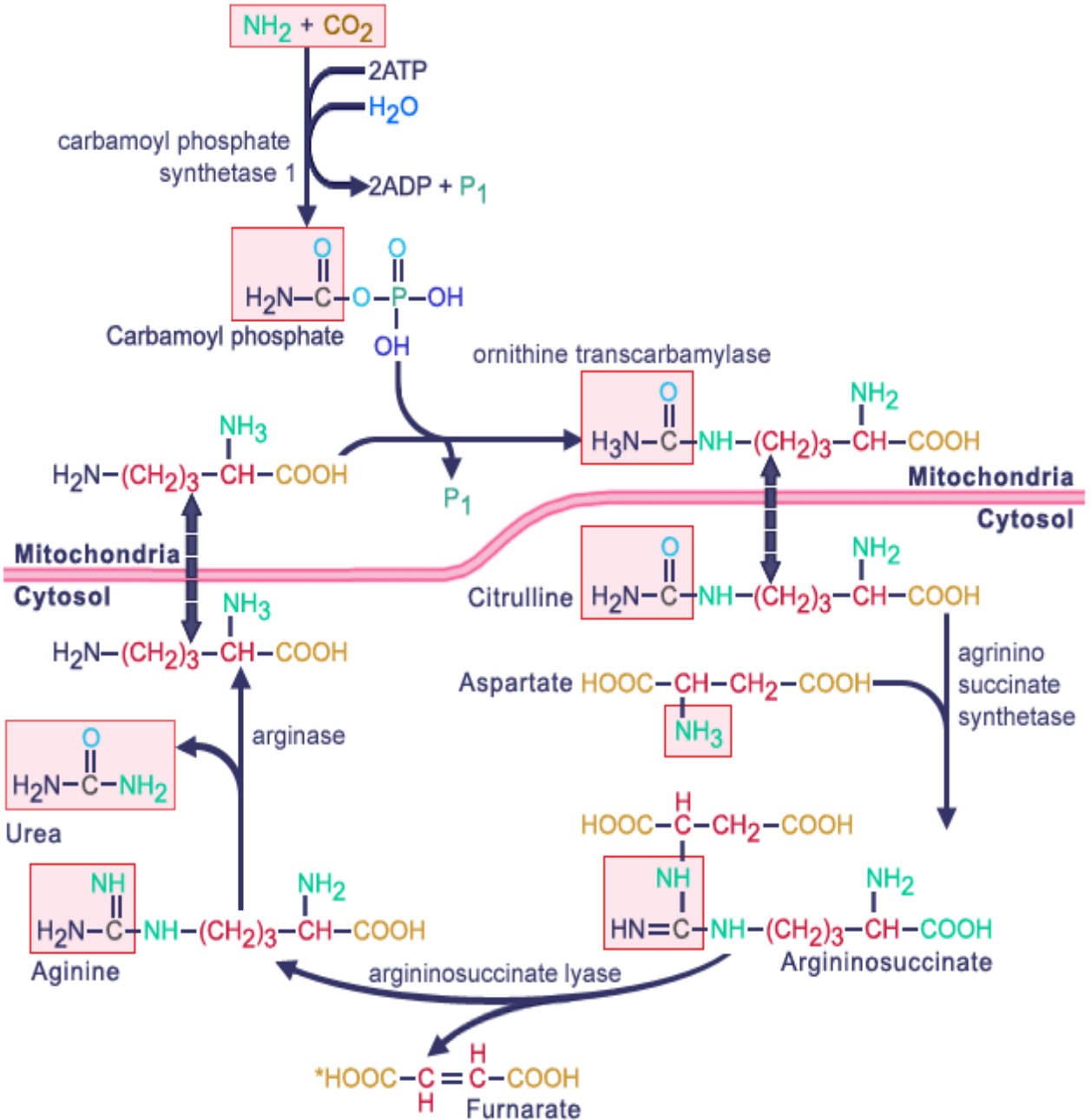
This is done by linking of aspartic acid and citrulline in the presence of argininosuccinate synthetase .

4-Cleavage of argininosuccinate into arginine and fumarate in the presence of argininosuccinase .

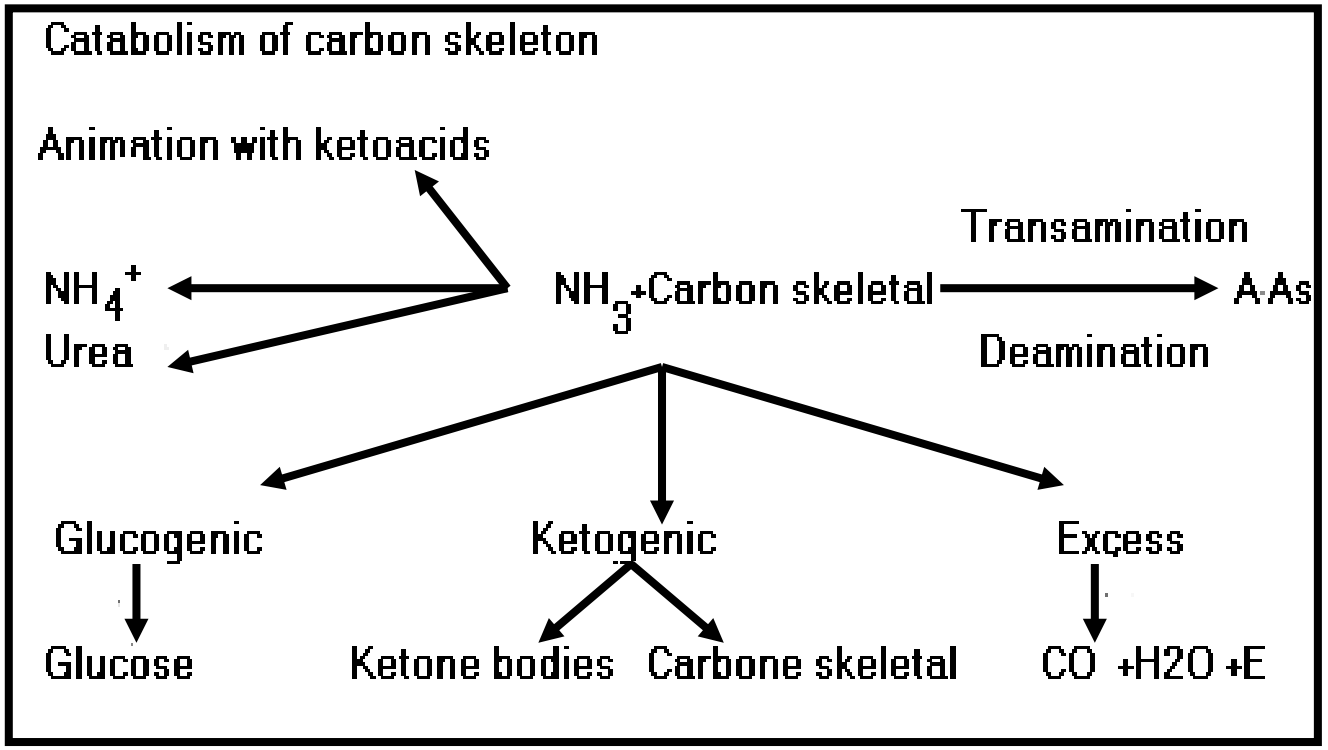
5-Cleavage of arginine into ornithine and urea in the presence of arginase.

This reaction completes urea cycle and regenerates ornithine as a substrate for reaction number(2) of urea cycle .

- **These reactions can be illustrated by the following diagram**



2-Catabolism of carbon skeleton



The carbon skeleton of amino acids is converted either to pyruvate which is then converted to carbohydrate in the process of gluconeogenesis (glucogenic amino acids) or to acetylcoA which is then converted to fat or ketone bodies (ketogenic amino acids). The excess catabolizes into CO_2 +H_2O and energy.

C-CLINICAL CORRELATIONS

1-Metabolic disorders of urea cycle

Urea cycle in liver involves (5) reactions catalyzed by(5) different enzymes, metabolic disorders associated with the deficiency of each of these (5) enzymes are known. The only function of urea cycle is to convert the toxic substance ammonia into non-toxic substance urea. All disorders of urea cycle cause ammonia toxicities which is more severe when the metabolic block occurs at reactions (1) or (2).

Clinical symptoms associated with all disorders of urea cycle are:

a-Intolerance to high protein diet.

b-Mental disorders.

c-Retard development of CNS.

d-Excessive urinary excretion of ammonia.

The principal disorders are :

1-Hyperammonemia type 1

There is deficiency of carbamoyl-p- synthetase.

2-Hyperammonemia type 2

There is deficiency of ornithine transcarbamoylase.

3-Citrullinemia

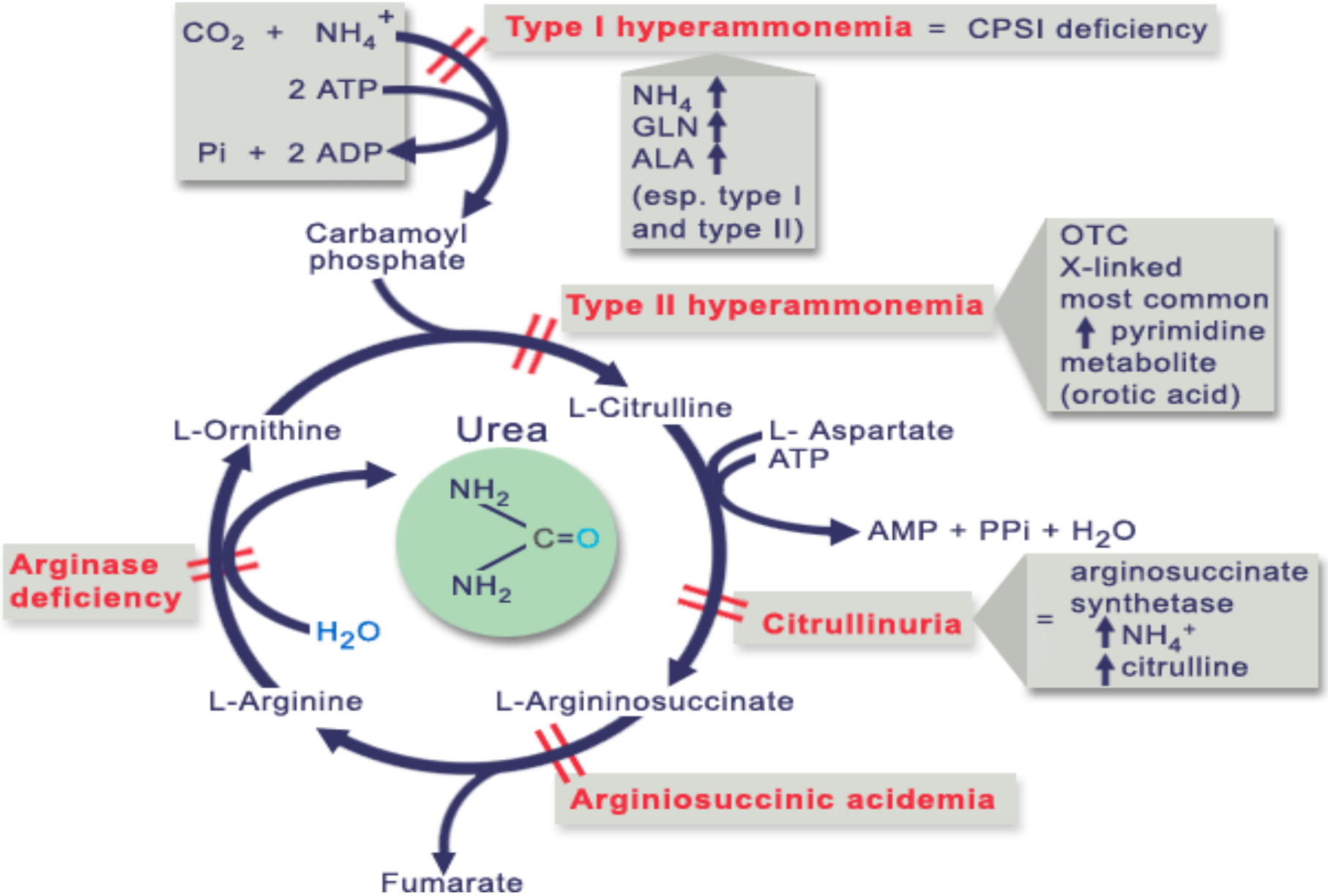
There is deficiency of argininosuccinate synthetase and characterized by: excessive urinary excretion of citrulline, high plasma and C.S.F. levels of citrulline .

4-Argininosuccinic aciduria

There is deficiency of argininosuccinase, characterized by elevated blood and C.S.F. levels of argininosuccinic acid.

5-Hyperargeninemia

There is deficiency of arginase.


CO_2 + NH_4^+
2 ATP
Pi + 2 ADP
Type I hyperammonemia = CPSI deficiency
NH_4 ↑
GLN ↑
ALA ↑
(esp. type I and type II)
Carbamoyl phosphate
Type II hyperammonemia
OTC
X-linked
most common
↑ pyrimidine metabolite
(orotic acid)
L-Ornithine
Urea
L-Citrulline
L- Aspartate
ATP
AMP + PPi + H_2O
Arginase deficiency
H_2O
Citrullinuria
= arginosuccinate synthetase
↑ NH_4^+
↑ citrulline
L-Arginine
L-Argininosuccinate
Arginiosuccinic acidemia
Fumarate

2-Metabolic disorders of amino acid metabolism

Abnormal metabolism of amino acids results in a number of metabolic disorders which must be early diagnosed and treated otherwise they produce irreversible brain damage.

Biochemical investigations include urinary and serum estimation of amino acids by different methods such as chromatography.

The principal disorders are:

1-Glycine

a-Glycinuria:

This is a rare metabolic disorder of glycine metabolism. Serum glycine is normal, but there is an excessive urinary excretion of glycine which is due to a defect in tubular transport and reabsorption of glycine.

b-Primary hyperoxaluria:

Another metabolic disorder of glycine metabolism, characterized chemically by a continuous high urinary excretion of oxalate which is unrelated to the dietary intake of oxalate (exogenous), but its of endogenous origin and comes from glycine after its deamination and conversion into glyoxylate which oxidizes to oxalate.

2-Histidine

Histidinemia:

An inherited metabolic disorder of histidine metabolism due to the deficiency of histidase and characterized by elevated serum and urine levels of histidine and clinically there is retardation of speech.

3-Phenylalanine

Phenylketonuria (PKU):

An inherited metabolic disorder of phenylalanine metabolism. It's due to the deficiency of the enzyme phenylalanine mono-oxygenase (phenylalanine hydroxylase), so phenylalanine can not convert into tyrosine, and its metabolism takes an alternative pathway as follows:

PEDIATRICS

PHENYLKETONURIA (PKU)

WHAT IS IT?

Rare genetic disorder of the inability to breakdown **PHENYLALANINE** into **TYROSINE** that can lead to **brain & nerve damage** if untreated

CAUSES/RISK FACTORS

AUTOSOMAL RECESSIVE GENETIC DISORDER

Must inherit one gene from **BOTH** parents

- → Family history
- → Ancestors that come from same region

SYMPTOMS

- → **Fair hair and skin**
  - ► Tyrosine creates melanin
- → Eczema (particularly perineal area)
- → Musty body odor
  - ► Phenylketones in sweat & urine
- → Neurological dysfunction
  - ► Failure to thrive
  - ► Seizures
  - ► Extreme hyperactivity
  - ► Progressive retardation or intellectual disability
  - ► Behavioral & emotional issues (↓ dopamine & serotonin)



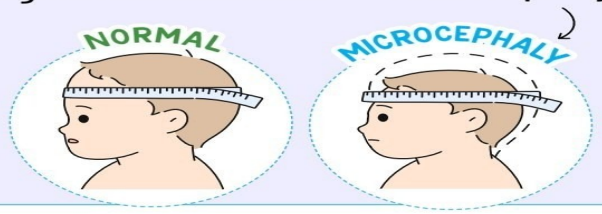
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PREGNANT MOTHERS WITH PKU

MUST FOLLOW PKU DIET DURING PREGNANCY

If not managed properly can cause:

- ► Miscarriage
- ► Cognitive disabilities
- ► Heart defects
- ► Microcephaly



PHENYLALANINE is an essential amino acid in the body that converts into **TYROSINE** through **phenylalanine hydroxylase**

↓

TYROSINE coverts into other products like:

- ► Dopamine
- ► Serotonin
- ► Norepinephrine
- ► Epinephrine

In **PKU** there is ↓ phenylalanine hydroxylase = impaired ability to use phenylalanine which leads to

BUILDUP OF PHENYLALANINE IN TISSUES + BLOOD

DIAGNOSTICS

TESTING REQUIRED IN MOST STATES

- → **GUTHRIE INHIBITION ASSAY**
  - ↳ Done when baby 24 hours old & after baby has ingested protein
- → **PLASMA PHENYLALANINE**
  - ↳ >4 MG/DL= PKU
- → **PRE-PREGNANCY SCREENING**

TREATMENT

- → Regular blood tests (more frequent when younger)
- → Sapropterin: ↓ phenylalanine

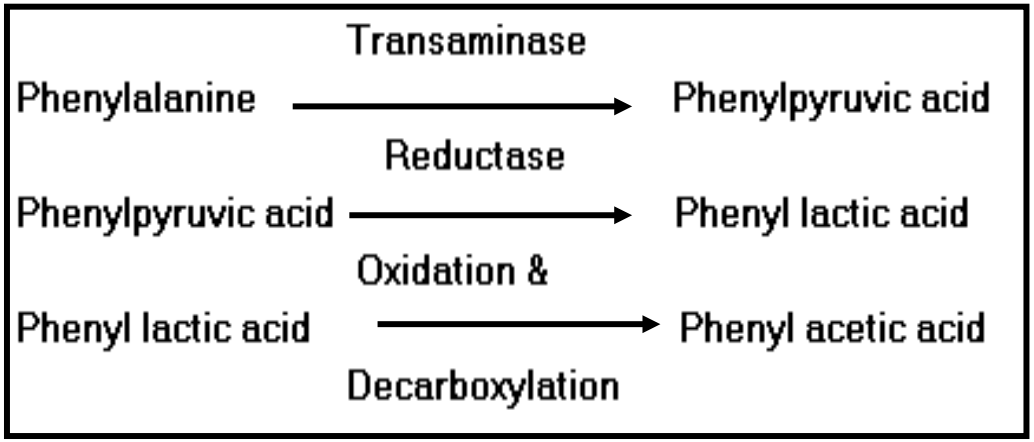
DIET

- → Phenylalanine-free formula (protein without phenylalanine)
- → **Limit protein intake** (protein contains phenylalanine)
- → Amino acid supplementation

FOODS TO MONITOR:

- ► Meat
- ► Dairy
- ► Fish
- ► Legumes

AVOID ASPARTAME! (artificial sugar)
Diet soda, sugar-free gum & foods, "lite" yogurt



So, there is urinary excretion of phenyl pyruvic, phenyl lactic and phenyl acetic acids and clinically there are mental retardation, vomiting, convulsion and special odor of urine (odor of rats).

4-Proline:

hyperprolinemia:- characterized by increased serum proline, caused by defective enzyme proline oxidase.

5-Tyrosine

a-Tyrosinemia (Richner –Hanhard syndrome):

Another metabolic disorder of tyrosine metabolism, there is an increase in the serum level of tyrosine, caused by a deficiency in the hepatic enzyme **tyrosine aminotransferase,** which leads to an accumulation of tyrosine in the blood and tissues. **excessive urinary excretion of tyrosine.**

The clinical symptoms are:

Hepatosplenomegally, liver cirrhosis, rickets, hyperphosphaturia and aminoaciduria.

b-Alkaptonuria:

Inherited metabolic disorder of tyrosine metabolism due to the deficiency of homogentisate oxidase a specific enzyme acts on homogentisic acid an intermediate compound formed during metabolism of tyrosine. Characterized by an excess urinary excretion of homogentisic acid, there are also darkening of urine when left in air, due to the oxidation of homogentisic acid into brownish – black pigment.

In addition, there is pigmentation of connective tissues, termed as Orchronosis-associated with characteristic arthritic changes that affect the large joints and spine in later half.

c-Albinism :

Another metabolic disorder of tyrosine metabolism due to the deficiency of melanocyte tyrosine hydroxylase resulting in hypomelanosis .

There are two principal types of albinism:

1-Oculocutaneous albinism (OCA) : melanin pigment (photoprotective pigment) is missing in the skin, hair and eyes.

2-Ocular albinism (OA): melanin pigment is missing mainly in the eyes .

6-Branched chain amino acids (valine, leucine and iso-leucine)

Maple syrup urine disease (branched chain ketonuria):

An inherited metabolic disorder in which the odor of urine resembles that of maple syrup or burnt sugar.

Characterized by elevated serum levels of valine, leucine, iso-leucine and their keto acids, it's due to the deficiency of the enzyme alpha keto acid decarboxylase which converts these amino acids into CO2 and acylcoA .

7-Tryptophan:

Hartnups disease:

Inherited metabolic disorder of tryptophan metabolism due to the deficiency of tryptophan di-oxygenase. Characterized by excessive urinary excretion of tryptophan, mental retardation and pellagra-like skin conditions.

The treatment supplementing the diet with additional amount of niacin, to alleviate the dermatological and neurological lesions.

8-Cysteine and methionine:

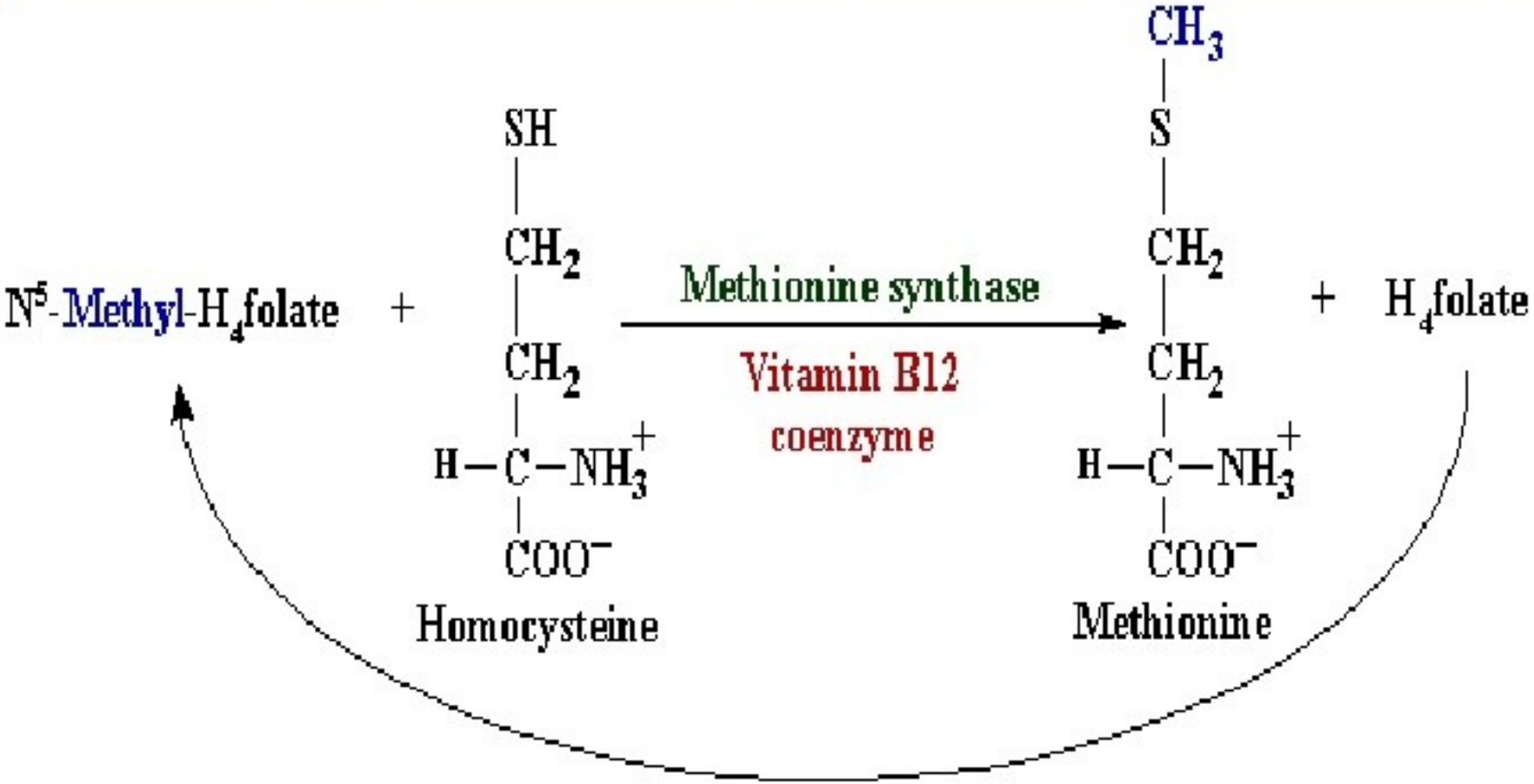
Homocysteinuria:- characterized by high plasma homocysteine and methionine and low levels of cysteine, the most common cause of homocysteinuria is defect in the enzyme cystathionine synthase which converts homocysteine to cystathionine.

Symptoms:

Are ectopia (displacement of the lens of the eye),

skeletal abnormalities, premature arterial disease, osteoporosis and mental retardation.

Methionine synthesis

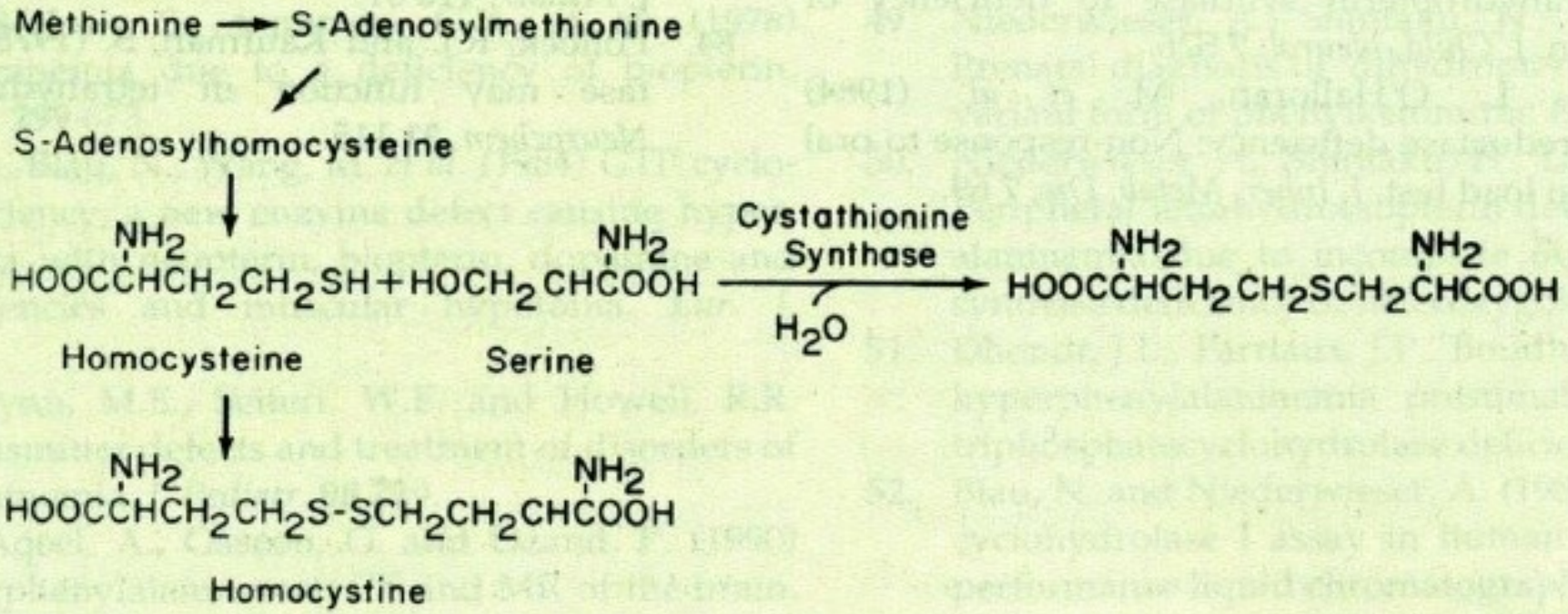


H_4folate accepts methyl groups in a number of different reactions an is converted back to N^5-Methyl-H_4folate

Homocystinuria

MAJOR PHENOTYPIC EXPRESSION

Ectopia lentis, vascular occlusive disease, malar flush, osteoporosis, accumulation of homocystine and methionine and defective activity of cystathionine synthase.



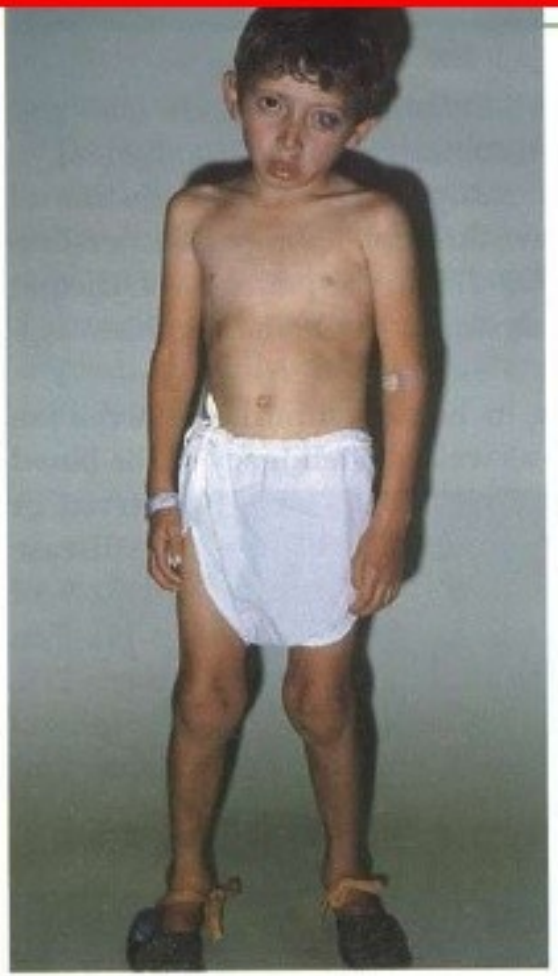


Fig. 21.2 M.G., a 6-year-old boy with homocystinuria. He had short stature and genu valgum.



Fig. 21.3 Closer view illustrates M.G.'s eyes. Subluxed lenses had previously been removed bilaterally, after which he developed glaucoma in the left eye. He had fair skin and hair and a pronounced malar flush.

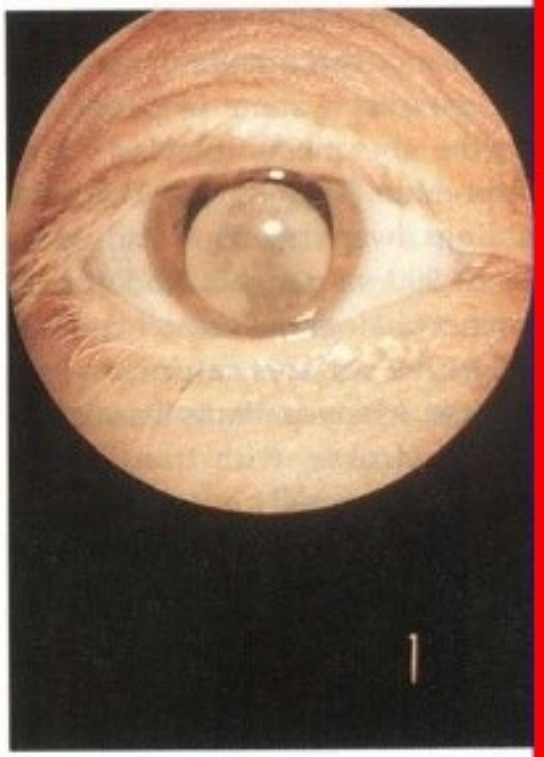


Fig. 21.4 The dislocated lens in homocystinuria is usually downward, while in Marfan syndrome it is upward.

3-Protein-calorie malnutrition

Inadequate intake of protein and/or energy may be observed. Affected individuals show a variety of symptoms, including a depressed immune system with reduced ability to resist infection. Death from secondary infection is common.

there are two extreme forms of malnutrition are;

Kwashiorkor and marasmus

1. Kwahiorkor:

Occurs when there is deficiency of protein only, is frequently seen in children after weaning at about one year of age, when their diet consists predominantly of carbohydrates. Typical symptoms include stunted growth, edema, skin lesions, depigmented hair, anorexia, enlarged fatty liver, and decreased plasma albumin concentration.

2.Marasmus:

Occurs in children younger than one year of age when the mothers breast milk is supplemented with thin watery gruels of native cereals, which are usually deficient is occurs when there is deficiency of dietary protein and calories. Typical symptoms include arrested growth, extreme muscle wasting, weakness, and anemia. Victims of marasmus do no show the edema or change in plasma proteins observed in kwashiorkor.

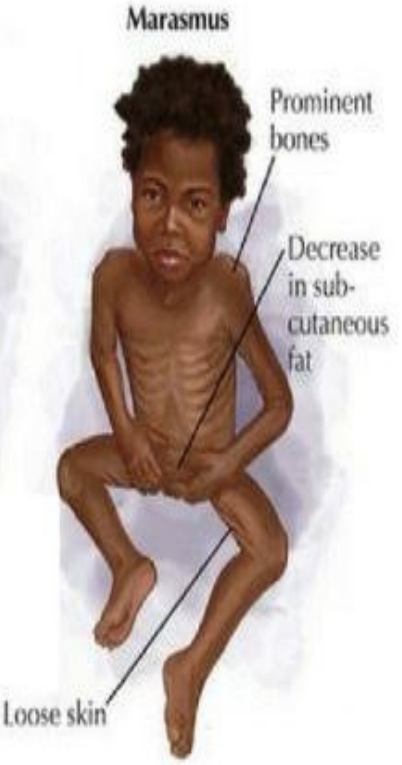
10 Differences between Kwashiorkor and Marasmus

www.majordifferences.com



Comparison Table

Kwashiorkor	Marasmus
It develops in children whose diets are deficient of protein.	It is due to deficiency of proteins and calories.
It occurs in children between 6 months and 3 years of age.	It is common in infants under 1 year of age.
Subcutaneous fat is preserved.	Subcutaneous fat is not preserved.
Oedema is present.	Oedema is absent
Enlarged fatty liver.	No fatty liver.
Ribs are not very prominent.	Ribs become very prominent.
Lethargic	Alert and irritable.
Muscle wasting mild or absent.	Severe muscle wasting
Poor appetite.	Voracious feeder.
The person suffering from Kwashiorkor needs adequate amounts of proteins.	The person suffering from Marasmus needs adequate amount of protein, fats and carbohydrates.



Kwashiorkor vs Marasmus

Acute phase proteins (APPs)

Acute phase proteins(APPs):-

The acute phase proteins are a group of proteins those plasma concentration changes in response to a variety of inflammatory states including infection, surgery, trauma, myocardial infarction, malignancy, the birth process, and any condition associated with tissue necrosis.

Some of these proteins are called positive acute phase proteins, meaning that their plasma levels increase, whereas others are designated negative acute phase proteins to indicate that their levels decrease.

Interleukins(IL), especially interleukin-1(IL-1), interleukin-6 (IL-6), tumor necrosis factor- α (TNF- α), and interferon-γ (IFN-γ), released by macrophages and lymphocytes, are primary agents which cause induction and release of these acute phase proteins will synthesize in the liver.

These acute phase proteins may be used to monitor progress the condition or its treatment.

Major acute phase proteins(APPs)

Positive acute phase proteins (+APPs)	Negative acute phase proteins (-APPs)
•Alpha-antitrypsin (AAT) •Alpha-anti chemotrypsin •C-Reactive protein (CRP) •Serum amyloid A protein •Haptoglobin •Ceruloplasmin •Fibrinogen •C3 •Hemopexin	•Albumin •Prealbumin •Retinol-binding protein •Transferrin (increase in late acute phase).

C-reactive protein (CRP)

Blood test measures the amount of CRP in blood, an indicator of inflammation in the body caused by infections, autoimmune disorders, or other chronic conditions. High levels, often linked to conditions like bacterial infections, inflammatory bowel disease, or autoimmune disorders such as lupus, suggest an inflammatory process is occurring.

A special type, CRP-hs, is used to assess long-term cardiovascular risk.

Recent research suggests that patients with elevated basal levels of CRP are at an increased risk of diabetes, hypertension, atherogenesis, and cardiovascular disease.

The researchers indicated that CRP may be only a moderate risk factor for cardiovascular disease.

Alpha anti trypsin (AAT):-

Is otherwise called alpha-anti protease or protease inhibitor(Pi). It inhibits all serine proteases (proteolytic enzymes having a serine in their active center). Such as plasmin, thrombin, trypsin, chemotrypsin, elastase, and cathepsin.

Binding of this inhibitor to protease is very tight; once bound it is not released.

Normally, about 95% of the anti protease activity in plasma is due to AAT.

Alpha anti trypsin is one of the plasma proteins, that inhibits activity of proteases especially elastase, which degrades elastin, a protein that gives elasticity to the lungs, so in normal individuals the activity of elastase is regulated by AAT. A genetic deficiency in AAT can lead to emphysema (lung disorder).

Clinical correlation

Excessive cigarette smoking leads to emphysema. As cigarette smoke inhibits the activity of AAT. Because the methionine residue at 385 position of AAT is important in the enzyme binding. This methionine may be oxidized to methionine sulphoxide by smoking. So emphysema is very common in smokers with normal AAT level and smoking will worsen the situation in AAT deficiency.

Cigarette smoke increases the number of neutrophils in the lung, and, therefore, increases the amount of elastase (elastase is released from neutrophils in lungs). Elastase then causes the tissue breakdown and loss of elasticity in the lungs which called emphysema.